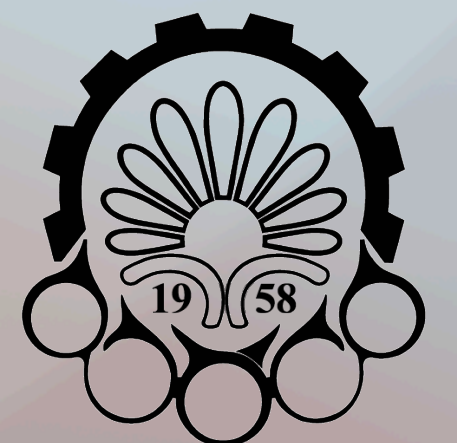


Standing on the Shoulders of Giants: Transfer Learning & Using Pre-trained Models

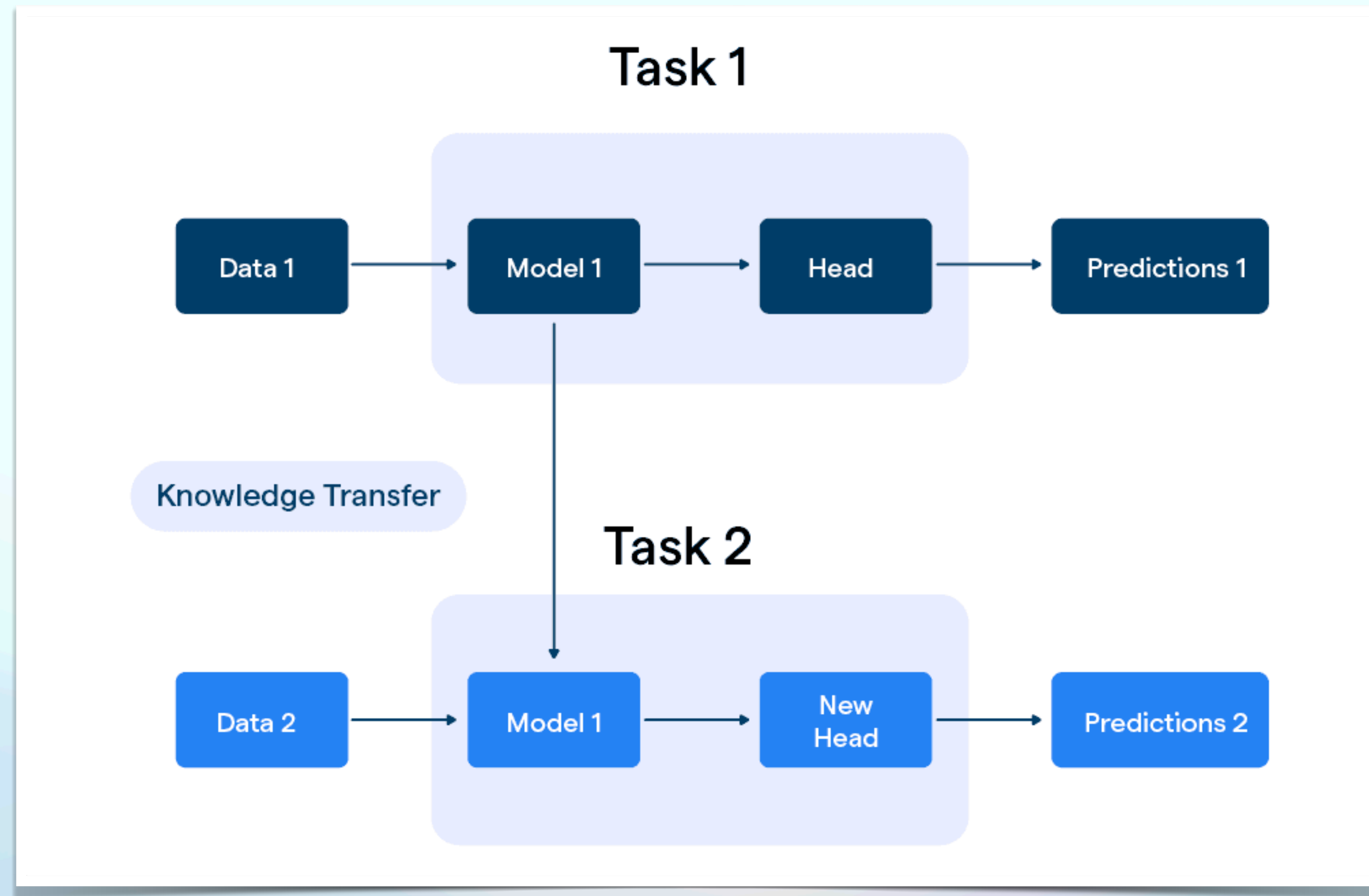
Computation Intelligence and its Application in Mechatronics

Spring 2025



Amirkabir University of Technology
(Tehran Polytechnic)

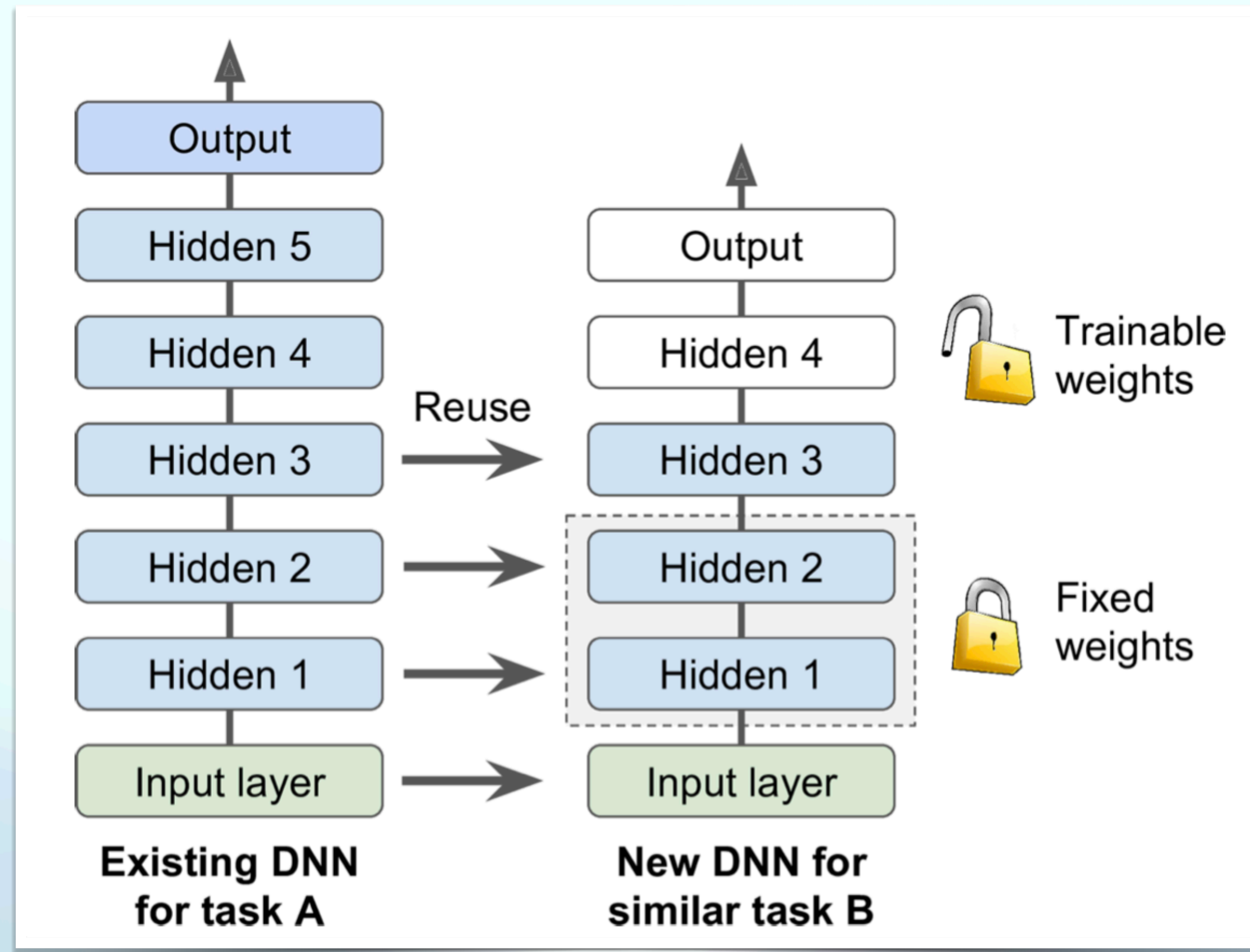
What is transfer learning?



Key Concepts

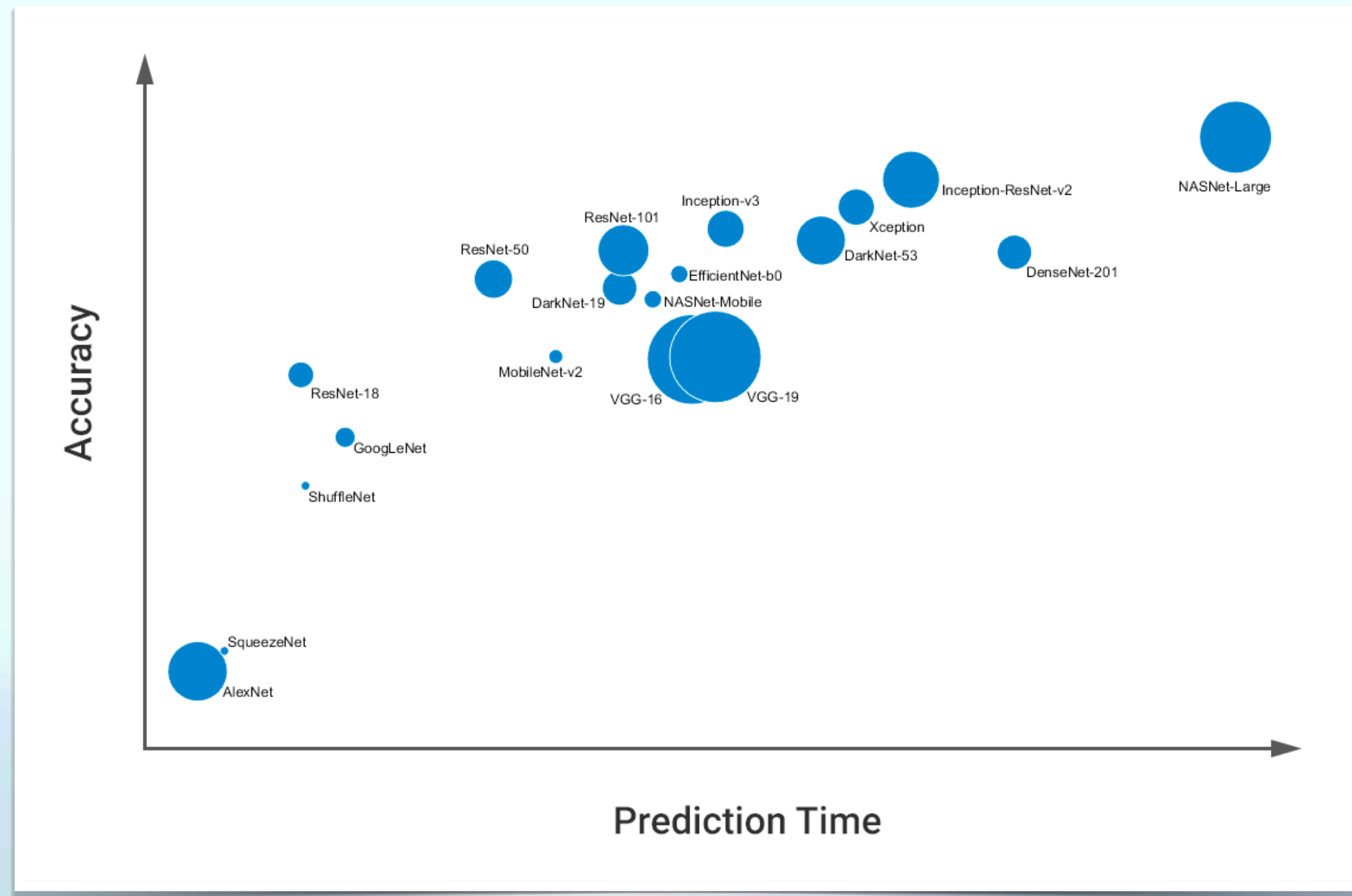
- **Pre-trained Model:** A model trained on a large dataset (e.g., ImageNet)
- **Fine-tuning:** Retraining some layers of the pre-trained model on the new task.

Reusing pretrained layers



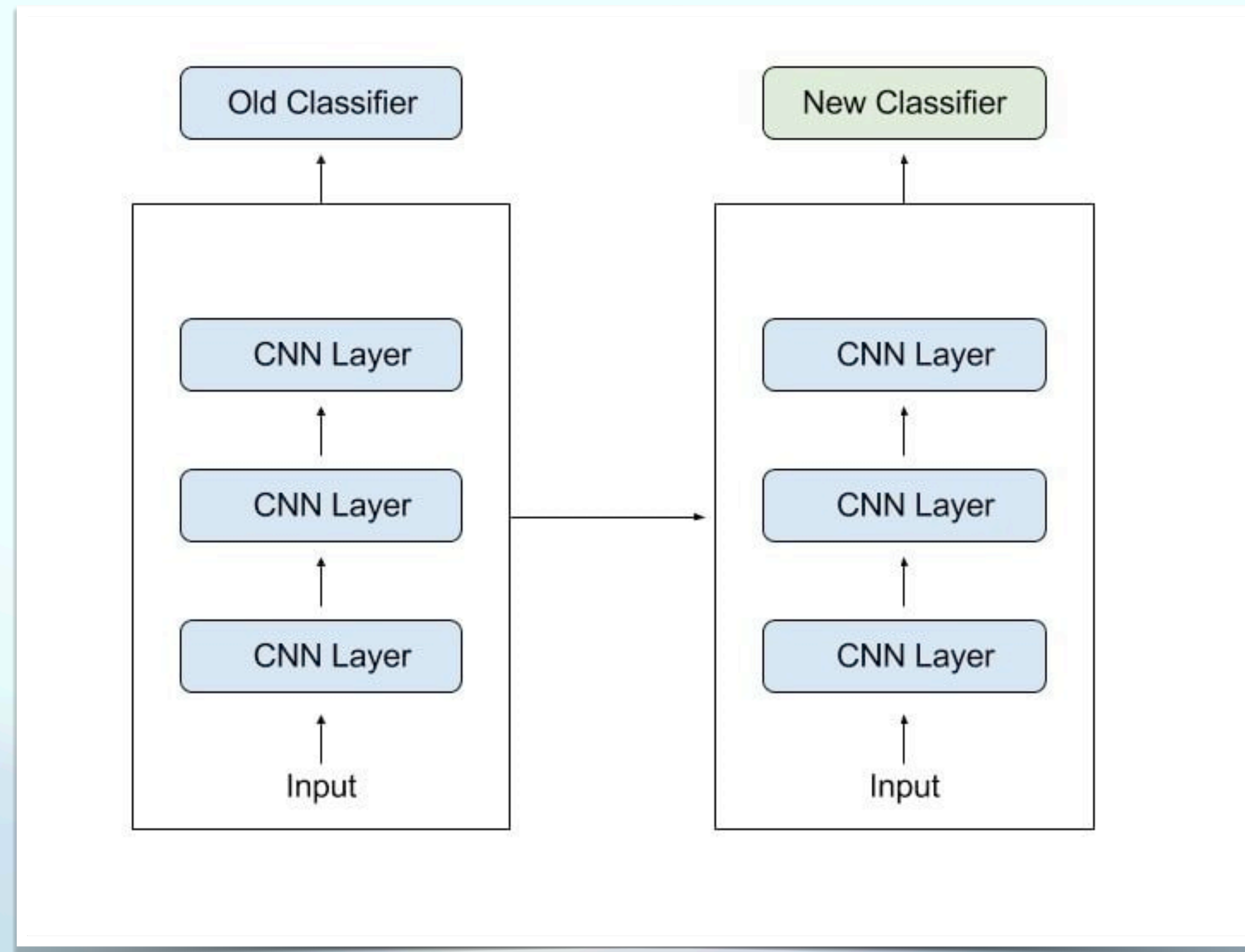
Pretrained Models for Transfer Learning

CNN pretrained models



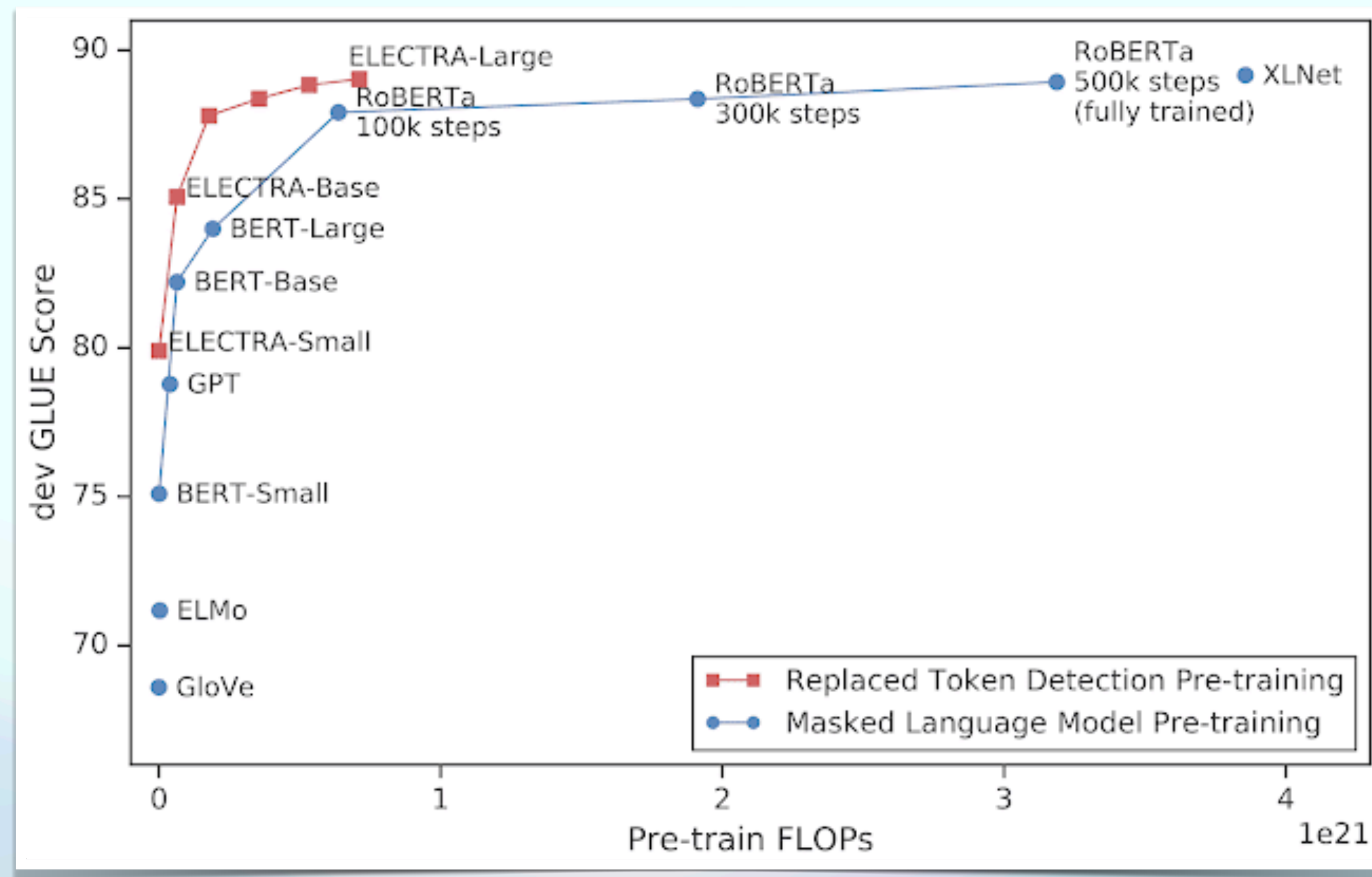
Pretrained Models for Transfer Learning

CNN pretrained models



Pretrained Models for Transfer Learning

NLP pretrained models



Transfer Learning Strategies

Strategy 1: Use pretrained model as a fixed feature extractor (freeze all weights)

Strategy 2: Unfreeze some top layers and fine-tune on new data

Strategy 3: Replace the top (output) layer to match your new task

Transfer Learning With Keras

Available via tensorflow.keras.applications

Steps:

Step 0- Preprocessing our data

Step 1- Loading Pre-Trained Model

Step 2- Freezing Base Layers

Step 3- Adding New Classification Layers

Step 4- Compiling and training the model